Shape of cubics

1 Yes

2

a they become very large in the positive direction

b they become very large in the negative direction

3 Odd, because the ends of the function point in opposite directions.

4

a $x < 0$

b $x > 0$

c $(0, 0)$

d The point of inflection

5

a $x > -3$

b $x < -3$

c $(-3, 0)$

6 y increases more slowly on $y = \frac{1}{2}x^3$ than on $y = x^3$

7 $x = -1, \quad x \geq 3$

8

a $(0, 0)$

b $(0, 3)$

c $(0, 4)$

Finding intercepts and equations of cubics

9

a Negative

b $(0, 3)$

c $x < 0$

d $x > 0$

e $(0, 3)$

10

a No

b No

c No

d Yes

11

a $x = -2$

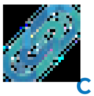
b $y = 4$

12 3

13

a negative

b $(0, 3)$



c **A**

14 $y = (x - 10)^3$

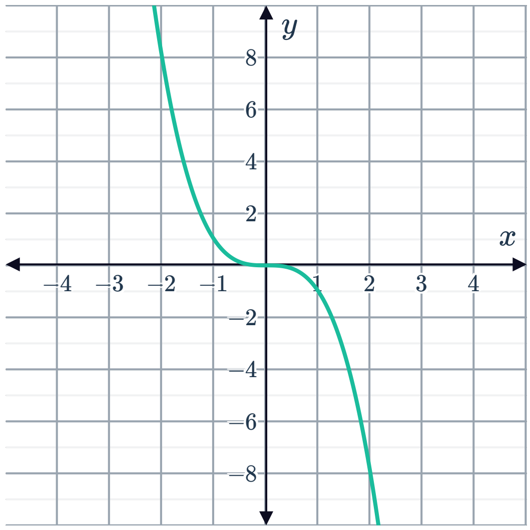
Graph of cubics

15

a

x	-2	-1	0	1	2
y	8	1	0	-1	-8

b



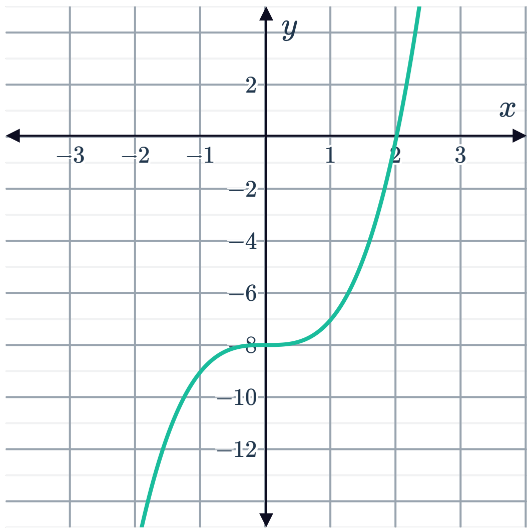
16

a $x = 2$

b $y = -8$

c $(0, -8)$

d



17

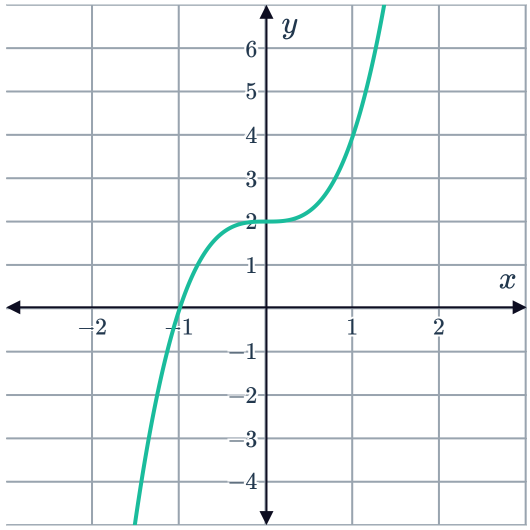
a

i Increasing

ii More steep

iii $(0, 2)$

iv



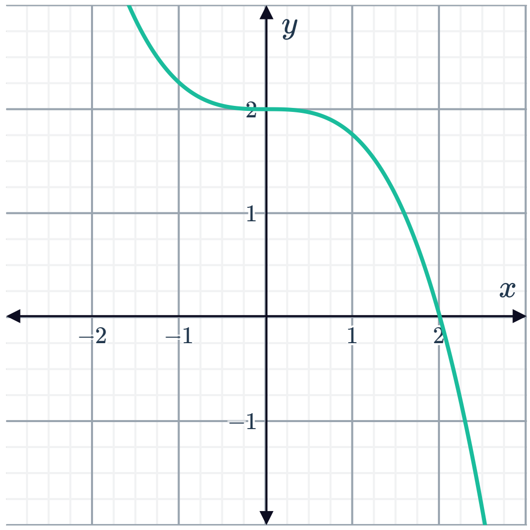
b

i Decreasing

ii Less steep

iii $(0, 2)$

iv



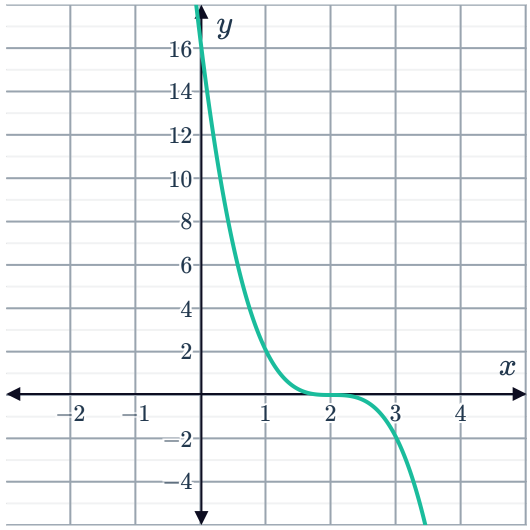
c

i Decreasing

ii More steep

iii $(2, 0)$

iv



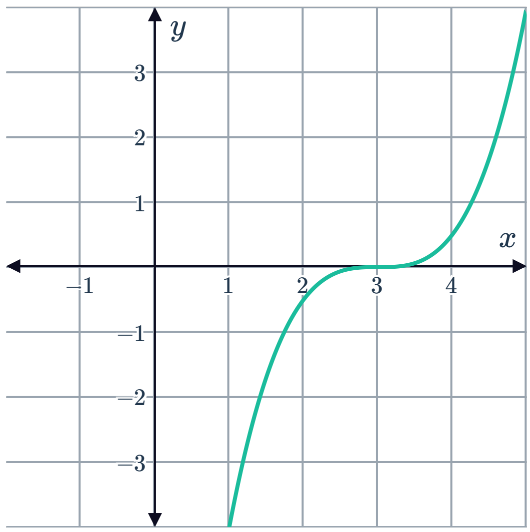
d

i Increasing

ii Less steep

iii $(3, 0)$

iv



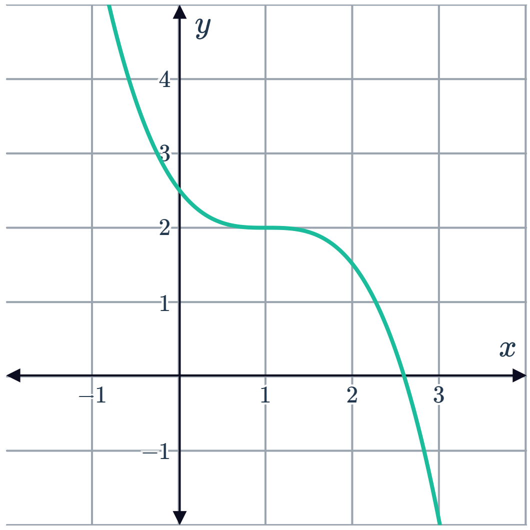
e

i Decreasing

ii Less steep

iii $(1, 2)$

iv



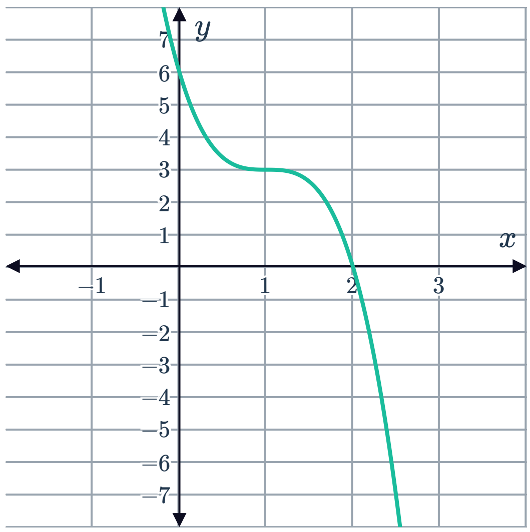
18

a $x = 2$

b $y = 6$

c $(1, 3)$

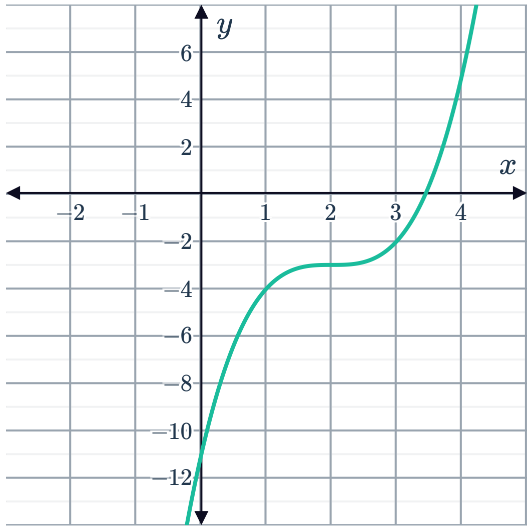
d



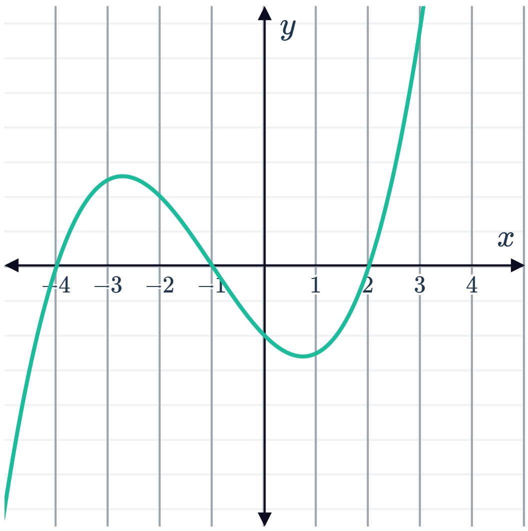
19

a Move the graph to the right by 2 units and down by 3 units.

b



20



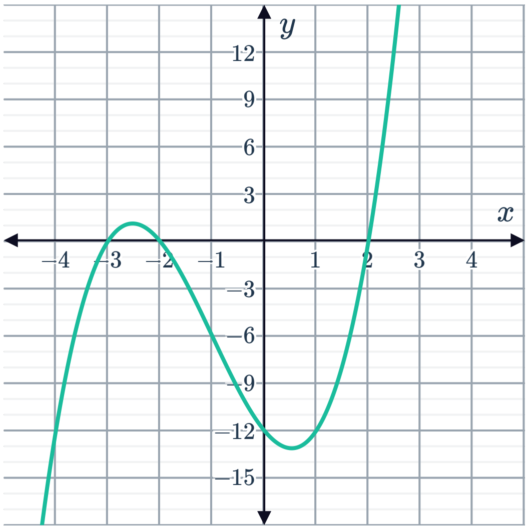
21

a

i $x=-3,-2,2$

ii $y=-12$

iii

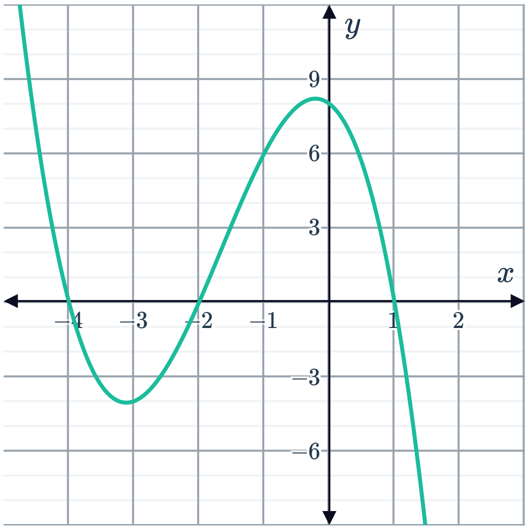


b

i $x=-4,-2,1$

ii $y=8$

iii



22

a $y=-x\left(x-3\right)\left(x+1\right)$

b $y=0$

c $x=0,3,-1$

d